The Effects of Greening Unconventional Monetary Policy

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March 2026

ABSTRACT

We bring evidence to the positive role played by central banks in supporting policies towards greening the financial system. We focus on the recent ECB *Monetary Policy Strategy Review*, which unexpectedly dedicated a whole workstream to climate change, and study its effect on the pricing and issuance of green bonds in the Eurozone. We find that Euro-denominated green bonds' Yield-to-Maturity decreased by a sizable amount of 12 bps following the announcement, and that the effect is concentrated among the issuers which are not the largest carbon emitters. Then, we measure changes in green bond issuance in the Eurozone compared to other jurisdictions. We find a significant increase in the amount of green bond issued in the Eurozone and in particular on the segment of ECB-eligible green bonds. We interpret the results as supportive of the role that the ECB and more generally central banks can provide in support of policies for greening the financial system. These findings are important for policymakers and central banks working on "greening" their monetary policy operations.

Keywords: Climate Change, Central Banks, Green Bonds, Carbon Emissions, Quantitative Easing, Monetary Policy, ESG

JEL classification: Q58, E52, E58, G12

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[‡]We are grateful to Ruediger Fahlenbrach, Rajna Gibson Brandon and Fabio Trojani. We have also received helpful comments from Ella D.S. Patelli, Andreas Fuster, Natalia Rostova and Erwan Morellec. Remaining mistakes are our own.

1 Introduction

In response to the surging understanding that climate risks ought to be addressed, financial institutions such as central banks have started exploring options for policy responses. The European Central Bank (ECB) is leading by example to support the development of green finance. As a large anchor investor for debt capital markets (Liebich et al., 2020), it has the ability to serve as a catalyst to direct financial flows to green priorities.

Various tools could be altered to conduct this policy "greening". However, the ones that are probably the most direct and visible are the Asset Purchase Programmes (APP) and, in particular, the Corporate Sector Purchase Programme (CSPP), which targets the corporate bond market in the European Union (EU). We focus in this paper on the effects of the unveiling of the *Monetary Policy Strategy Review*, on July 8, 2021. On that day, ECB officials have announced (henceforth, "ECB announcement") their action plan to incorporate climate criteria into their monetary policy operations, with the ultimate goal of redirecting funds towards green finance[1].

However, this concept remains largely undefined in terms of instruments. As a result, several uncertainties persist about the implementation details of the "green shift". On the one hand, green bonds can be expected to play a preferred role as their proceeds finance climate-friendly projects[2], and they were shown to be an effective instrument to signal and effectively carry out a company's effort towards carbon emissions reduction (Flammer, 2021). On the other hand, the market, while flourishing, remains small and juvenile, in particular due to the lack of oversight and standardization (Deschryver and de Mariz, 2020), despite some recent initiatives carried by the European Commission, such as the EU Taxonomy and the EU Green Bond Standard. The ECB could also exclude specific firms and/or sectors based on climate criteria and thus reduce the overall carbon footprint of its CSPP portfolio holding (Papoutsi et al., 2021), as proposed by Fahlenbrach and Jondeau (2021) for the case of the Swiss National Bank. However, pure divestment (for example without repurchase of green bonds in those sectors) would not help brown segments of the economy do their transition to sustainable practices.
Both potential approaches contrast with the current implementation of the CSPP, which since its inception in 2016 has been based on sectoral amounts issued, an approach often referred to as "Market Neutral". "Market neutrality" has an important drawback: as the ECB buys bonds in proportion to the amount of bonds outstanding, sectors where high share of capital is funded by bonds are over-represented, leaving the ECB portfolio tilted towards high emission sectors (Papoutsi et al., 2021). Under the "Market neutrality" principle, green bonds are not excluded from the CSPP, but bought in proportion to the market value of outstanding bonds, and turning a blind eye to their "greenness". In opposition to this

[1] https://www.ecb.europa.eu/press/pr/date/2021/html/ecb.pr210708_1~f104919225.en.html

[2] https://www.icmagroup.org/sustainable-finance/the-principles-guidelines-and-handbooks/green-bond-principles-gbp/

"Market neutrality" paradigm, Executive Board Member Isabel Schnabel has notably argued in favour of a "Market efficiency" approach, which *"would explicitly recognise that a supposedly "neutral" market allocation may be suboptimal in the presence of externalities"*[3].

In this paper we study whether the ECB announcement has effectively been interpreted by market investors as a signal of its increased awareness of its environmental impact, and of its commitment to play a major role towards greening the financial system. In particular, we focus on the effects in the pricing and issuance of green bonds in the Eurozone. Understanding the effects of the ECB announcement on the green bond market is particularly important as numerous central banks have expressed their interest in supporting policies to scale up green finance (Dikau and Volz, 2021), but also given that badly planned climate targets could backfire and harm central banks' reputation (Hansen, 2021).

In line with a large body of literature studying the effects of ECB asset purchase programmes (Bremus et al., 2021; Grosse-Rueschkamp et al., 2019; Todorov, 2020; Koijen et al., 2021), we find strong evidence of the role of the ECB in the Eurozone green bond market. When studying the pricing effects of the ECB announcement, we find that EUR-denominated green bonds Yield-to-Maturities decreased by a sizable amount of 12 bps in secondary market transactions, when compared to non-Eurozone green bonds. The effect is concentrated in green bonds issued by firms which are not the highest carbon emitters. We next move to the question of market reaction of conventional bonds, depending on the greenness of their issuer. We find that conventional bonds issued by the highest carbon emitters reprice considerably following the ECB announcement. On the contrary, we do not find a significant pricing reaction when comparing conventional bonds issued by top and bottom carbon emitters in each sector. We interpret these effects on the conventional bond market as suggestive of market investors remaining cautious regarding the ECB paradigm shift in its APP.
Building on previous literature (Fama and French, 1993; Chen et al., 2014; Chordia et al., 2017) on market segmentation between equity and bond markets, we study the reaction of equity markets following the ECB announcement. Equity markets have experienced a strong growth in ESG and sustainable related investments. Anecdotal evidence from the United States Forum for Sustainable and Responsible Investment (USSIF) suggests that in 2021 a total of USD 17.1 trillion are invested according to sustainable and responsible investment principles (SRI) (Gibson Brandon et al., 2021). This large inflow of funds towards SRI mandates are linked to flow driven price reaction on the underlying assets. In the context of our analysis, we test whether equity market investors react positively to the ECB announcement. We find that when sorting Stoxx600 firms by their carbon emissions, a portfolio which is long high carbon emitters and short low carbon emitters earn a positive and significant abnormal return in the day following the ECB announcement, while issuers of ECB-eligible green bonds do not experience such a positive reaction. Our previous findings show that the ECB announcement did not alter the cost of debt financing for green

[3] https://www.ecb.europa.eu/press/key/date/2021/html/ecb.sp210614~162bd7c253.en.html

firms compared to brown firms. Instead, we find a positive stock reaction of those green firms to the ECB announcement, which we interpret as due to ESG-driven investors in the equity market.

Motivated by our first findings, we study the change in green bond issuance behaviour of firms in the Eurozone following the change in secondary market pricing for green bonds. First, we compare the green bond issuers' response in the Eurozone with other jurisdictions and find that Eurozone green bond issuers substantially increase green bond issuance following the ECB announcement, with the effect being more pronounced for banks. This finding is in line with the banking literature documenting that banks respond faster to changes in financing conditions in the bond markets. Second, we examine the effect of the ECB announcement within the Eurozone, by comparing the issuance behaviour between ECB-eligible vis-à-vis ECB-ineligible firms. We find that issuance of ECB-eligible green bonds increases compared to non-eligible green bonds, with the effect being stronger within the non-banking sector.

In summary, our findings provide evidence of the positive effect of the ECB-announcement of inclusion of green considerations within its monetary policy operations. They are important for central banks when assessing the potential impact of their commitment to supporting green financing instrument within their monetary policy operations.

This paper speaks to several strands of literature.

First, our paper is related to the literature studying the effects of asset purchase programmes that target corporations. De Santis and Zaghini (2019) find that direct corporate bond purchases by the ECB decrease all (i.e. for the most part conventional) eligible bond yields. They also find that, spreads have decreased not only for eligible bonds, but also to a lesser extent for non-eligible bonds, due to some spill-over effects. Bremus et al. (2021) analyse the effects of the same programme, but focus on green bonds. They find that the yields of eligible green bonds have significantly declined compared to non-eligible green bonds. In addition to price effects, Todorov (2020) studies the impact of large-scale bond purchases on corporate debt issuance, and finds that there was an increase both in the absolute number and in the notional amount of newly issued CSPP-eligible bonds after the CSPP announcement. He also finds that the effect is more pronounced for credit-constrained firms, which benefit the most from the decreased cost of debt financing. Grosse-Rueschkamp et al. (2019) further show that cheaper bond financing provided by the central banks' purchases led to a decrease in loan demand. They document that asset purchase programmes transmitted to the real economy via banks, which are faced with smaller loan demand from bond issuing firms, were able to provide additional lending to firms.

Second, given the nature of the event we focus on (the "green shift"), which represents a demand shock towards sustainable assets, our paper is also related to literature studying demand for this particular type of assets. Pastor et al. (2020) provide a theoretical framework in which heterogeneous investors have tastes for green assets. In equilibrium, asset prices are affected by the dispersion of green investors in the economy and expected returns arise from a two-factor model in which green assets have negative loadings on the ESG factors (i.e. in equilibrium expected returns in green assets are lower because they provide hedge against undiversifiable climate risk). However, unexpected shifts in demand for green assets positively impact equilibrium expected returns and green assets can still outperform brown assets. In line with this theoretical insight, a green shift in the asset purchase programmes represents a sizable demand shift in the Eurosystem financial market and should be incorporated in asset prices in equilibrium.

In particular, our findings are close to the literature studying the role of institutional investors in influencing greener policies for their portfolio firms. The literature has mostly focused on two main channels used by institutional investors to express their views in relation to green policies: the "Voice" and the "Exit" channel. Azar et al. (2020) provide evidence on the effects of the "Big Three" institutional investors in incentivizing carbon reduction of their portfolio firms. They find that engagement by large institutional investors such as Blackrock, Vanguard and State Street Global Advisors leads larger polluting firms in their portfolio holdings to reduce carbon emissions. Gibson Brandon et al. (2021) focus on the sustainable and responsible footprint of institutional investors globally. They show that raising demand from institutional investors with SRI mandates and subsequent to price-pressure on top ESG stock performers, has driven a significant link between portfolio-level sustainability at investor level and their risk-adjusted performance. We study whether central banks, which do not directly engage with corporations, but have wider communication channels, can have similar impact.

Third, it is related to the literature on corporate green bonds and green financing. Flammer (2021) shows that green bonds are an effective instrument to signal a firms' commitment to reduce carbon emissions. She also shows that green bonds do not attract better pricing compared to similar bonds. Nevertheless, the shift in demand for sustainable finance instruments from central banks could be a new catalyst for the development of sustainable related financing products. Baker et al. (2018) study the pricing and ownership patterns in the municipal green bond market in the US. While their focus is in the green premium when comparing green and conventional bonds, and whereas we focus on the pricing reaction within green bonds and conventional bond markets, their findings are useful to conceptualize the channels through which the ECB announcement could trigger a re-balancing between institutional investors.

2 Hypothesis Development

Hypothesis 1: Bond and Stock Market Reactions In the first part of the paper, we study whether the ECB "green shift" announcement led to a decrease in the cost of bond finance for green projects and firms. More precisely, we examine the effect of the announcement on the pricing of outstanding corporate green and conventional bonds. Our analysis builds on previous work that has focused on the creation of the CSPP, and has found that it led to a decrease in the Yield-to-Maturities of corporate bonds on secondary market transactions, for both conventional (Zaghini, 2017) and green bonds (Bremus et al., 2021).

First, we focus on green bonds, which are used to signal a firm's commitment to effectively reduce its carbon emissions (Flammer, 2021). In line with the findings of Bremus et al. (2021), we expect eligible green bonds to react by decreasing their Yield-to-Maturities compared to green bonds not affected by the ECB announcement. In addition, while green bonds are expected to play an important role going forward, they are still in a limited market. We extend our analysis to study conventional bond reaction depending on issuers' carbon emissions. Depending on the implementation of ECB "green shift", we evaluate the investor reaction with respect to the "Market Efficiency" and "Market Neutrality" paradigms. On the one hand, the ECB could tilt away from the most polluting companies implementing a global screening approach and thus depart from "Market Neutrality". Under this scenario we would expect the top polluting firms to negatively react. Alternatively, the ECB could tilt away from the most polluting companies at sector level by implementing a within-industry screening approach and maintain its "Market Neutrality". Under this scenario, we would see significant market reaction between sector-specific long-short portfolios sorted on emission levels.

Hypothesis 2: Green Bonds Adoption In the second part of the paper, we focus on the role of the ECB announcement in boosting the issuance of particular segments of the green bond market in the Eurozone.

We consider the announcement as an exogenous shock on the demand for green bonds and in particular for eligible green bonds. Higher demand for green bonds translates into lower Yield-to-Maturities for corporate green bonds and subsequent higher incentives for green bond issuers to increase issuance. The exogeneity comes from the unexpected importance given to climate change issues within the recently announced Monetary Policy Strategy Review of the ECB, as detailed in Section 3.2.

As shown by Koijen et al. (2021), ECB purchases of eligible securities are accompanied by lower liquidity and lower Yield-to-Maturity, and it is possible that investors rebalance their portfolio towards non-eligible segments of green bonds which offer relatively higher Yield-to-Maturity. Those spill-over effects documented for example by Bremus et al. (2021) in the case of CSPP creation reduces Yield-to-Maturity in non-eligible segments of bond markets. It is therefore ex-ante not clear whether non-eligible green bonds issuance should be subdued to eligible green bond issuance given the increased incentive for both segments of the green bond market.

3 Institutional Details

3.1 ECB Asset Purchase Programmes

Asset Purchase Programmes In the aftermath of the Great Financial Crisis and Sovereign Debt crisis in Eurozone, the ECB implemented a package of non-standard monetary policy measures, the Asset Purchase Programme (APP), with the objective of supporting the monetary policy transmission mechanism to ensure price stability, and of putting a halt to the persistently weak inflation dynamics. The APP consists of four separate programmes targeting specific segments on the debt markets: the Public Sectors Purchase Programme (PSPP), the Corporate Sectors Purchase Programme (CSPP), the Asset Backed Securities Purchase Programme (ABSPP) and Covered Bond Purchase Programme (CBPP)[4]. The APP followed the *Security Market Programme*, which included purchases of covered bonds and sovereign bonds. The APP as we intend here, started in January 2015 with the announcement of the extended asset purchases in order to restore inflation below but close the 2%. Initially the programme reflected the structure of the *Security Market Programme* by extending the ABS purchase program and the third covered bond purchase programme and introducing a structured programme for public debt program. Only in 2016, after persistent low inflation, the ECB decided to introduce the CSPP programme, targeting public corporate debt in the Eurozone.

Corporate Sector Purchase Programme The CSPP, announced in April 2016[5] and implemented from June 2016, specifically targets non-bank corporations established in the euro area, and aims at stimulating credit provision for those non-bank corporations with access to the bond market.
To be considered eligible under the CSPP, a debt instrument has to satisfy the following conditions:

- have a minimum rating of BBB- or equivalent,
- be denominated in euros,
- be issued by a non-credit corporation established in the euro area,
- have a remaining maturity of 6 months to 30 years at the time of purchase.

Purchases are, for now, guided by the overarching principle of "Market Neutrality" to avoid market distortions. In particular, it does not discriminate based on industry sector composition and purchases are based on outstanding amount issued in the eligible universe.
While there are no screening criteria based on environmental or social criteria, the ECB has already bought green bonds under the CSPP programme. As of the beginning of 2022, a total of 70 unique green bonds have been purchased within the CSPP, as detailed in Figure A.2a and Figure A.2b.

[4]For more details see directly the dedicated website on the ECB webpage: https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html
[5]https://www.ecb.europa.eu/press/pr/date/2016/html/pr160421_1.en.html

Third Covered Bonds Purchase Programme (CBPP3) While credit institutions, such as banks, do not meet the eligibility criteria for the CSPP, these financial institutions are directly affected by APP purchases under the Covered Bond Programmes. Following the extended purchases measures under the APP starting from 2014, the Third Covered Bond Purchase Programme[6] (CBPP3) was conducted between October 2014 and December 2018. The CBPP3 was meant to further strengthen the balancing channel of transmission of monetary policy with further easing of funding and credit conditions in the Eurozone. Given the role of banks in originating loans as collateral for covered bonds, it is expected that a sizable portion of banks' green bonds to be eligible for CBPP3 purchases thus contributing to the large issuing incentives for banks in the green bond markets.

3.2 The announcements

Since September 2020, ECB Executive Board Members have started to be increasingly vocal on the topic of climate change and on the possible role for central banks in the transition to a greener economy. As detailed in Figure 1, we focus on one particular event, the unveiling of the Monetary Policy Strategy Review on July 8, 2021. It was unexpected that climate issues would be given such a central role in this review (Reichlin et al., 2021), as ECB officials typically discuss those subjects in speeches that are not directly related to the conduct of monetary policy. This event was also an important one for the ECB, as it directly concerns its future strategy, and it attracted a lot of media coverage. It thus can be thought of as an important signal from the ECB on its intention to further incorporate climate change considerations in its future plan of action. While no concrete proposals were made on that day, a parallel proposal by the European Commission for an "European Green Bond Standard"[7](EUGBS) gave some credibility to the ECB announcement in favour of green assets.

4 Data

Sample Our benchmark sample is given by the constituents of the Stoxx (Europe) 600 index which represents large, mid and small capitalization companies across 17 countries of the European region. The choice of the benchmark sample is guided by several considerations. First, selecting the largest 600 publicly traded stocks in Europe allows to focus on firms with similar capital structure and in particular with access to debt markets. Second, among the 17 countries making up the index, there are some non-Eurozone ones, such as the United Kingdom, Switzerland and Sweden, which are not ECB-eligible and thus may constitute suitable controls. Finally, by focusing on this set of firms allows us to perform the event study analysis of stock reaction without having to account for factors related to liquidity in a first approximation.

[6] https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html#cbpp3
[7] https://ec.europa.eu/commission/presscorner/detail/en/ip_21_3405

Figure 1: Timeline of ECB announcements regarding the incorporation of climate change considerations into its monetary policy strategy. The full list of speeches is available at https://www.ecb.europa.eu/home/search/html/climate_change.en.html.

Data Sources Data on Stoxx 600 index constituents are from *Compustat Global* index dataset which includes security level information on the constituents. Security prices data are from *Compustat Global Securities Daily* dataset, and collected at a daily frequency. We retrieve yearly accounting data from *Compustat Global*. In addition, we obtain information on firms' greenhouse gas emissions ("GHG") from *Refinitiv Asset4*. This dataset contains yearly frequency and firm level information on Scope 1, Scope 2 and Scope 3 emissions.

Daily Mid Yield-To-Maturity for the period between July 2020 and end of December 2021 are obtained from *DataStream* using bond ISINs from *Bloomberg Fixed Income* database. Among those bonds, we identify the green ones through Bloomberg's label (more precisely, bonds for which the field "Green bond indicator" is "Yes"[8]). Following Flammer (2021), we exclude bonds whose issuer's BICS (Bloomberg Industry Classification System) is "Government".

Variable Definitions We use two approaches to classify green and brown firms in our sample using data on **direct** CO2 equivalent emissions in 2019 (i.e. **scope 1**), generated by burning fossil fuels and production processes. In the first one (the "global screening" approach), we compute the distribution of emissions in the entire sample of Stoxx600 firms (i.e. we fully ignore sectors), and use the bottom and top quartiles of this distribution to identify respectively green and brown firms. In the second one, we

[8]Bloomberg identifies Green Bonds following a proprietary methodology which screens debt documentation and labels as green debt securities those with identifiable information on green project or assets in the use of proceeds. For now, we do not have any information regarding the certification of those green bonds. https://data.bloomberglp.com/bnef/sites/4/2015/09/BNEF_Green-Bonds-Terminal-Guide_H2-2015-update.pdf

perform this procedure at each sector (defined through 4-digit GICS), in order to identify the green and brown firms in each of those sectors. The rest of the variable definitions are available in Table C.1 in the paper Appendix.

Descriptive Statistics Table 1 presents descriptive statistics on bond issuance by Stoxx600 firms. These firms are major actors in the corporate green bond market, as they represent half of the total amount issued in corporate EUR-denominated conventional bonds, and about a quarter of the total amount in corporate EUR-denominated green bonds. Comparing the proportion of total amount issued and the total number of corporate green bond issues we infer that Stoxx600 firms issue larger tickets, in line with the finding for conventional corporate bonds.

Table 2 presents a host of descriptive statistics for each of the subsets of corporate bonds under consideration: (i) conventional EUR-denominated bonds issued by Stoxx600 firms, (ii) green EUR-denominated bonds issued by Stoxx600 firms, and (iii) green SEK-denominated bonds. Conventional bonds have on average, in secondary market transactions, a higher Yield-to-Maturity than green bonds[9].

For our baseline sample of Stoxx600 firms, some firm-level descriptive statistics are presented in Appendix Table C.2. In Appendix Table C.3, we also estimate a correlation table for the same set of firms. Interestingly, the "# EUR Green Bonds issued" is significantly and positively correlated with both "Brown firm (global scope 1 emissions)" and "Brown firm (sectoral scope 1 emissions)" suggesting that brown firms either at global level and sectoral level are among the typical issuers of green bonds in our sample. This suggests that brown firms are among the type of firms fueling the growth in issuance of EUR-denominated corporate green bonds in our sample.

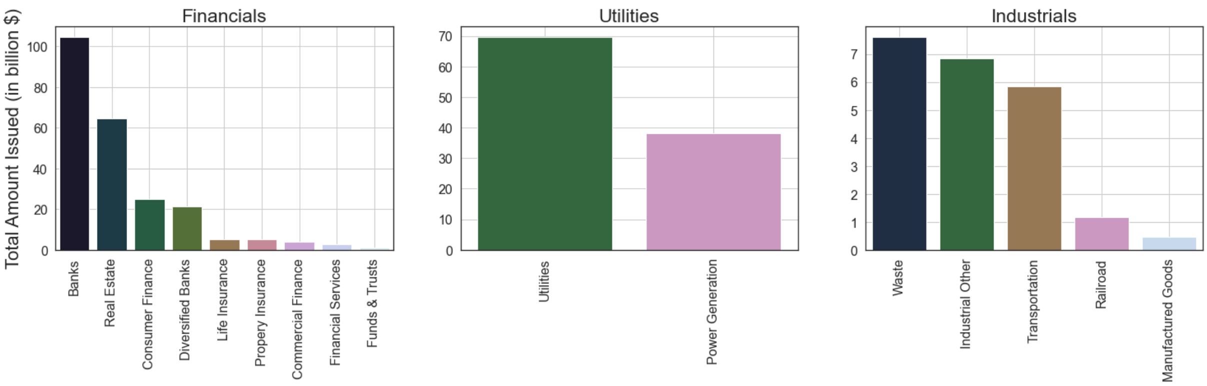


Figure 2: shows (from the left panel) the aggregate EUR-denominated green bond issuance in USD dollar billion at BICS Level 2 for the top 3 issuing sectors: Financials, Utility firms and Industrials.

Figure 2 shows the total amount of EUR-denominated corporate green bonds issued for the top 3 Bloomberg Industry Classification System Level 1 (BICS Level 1) issuing sectors ("Financials", "Utili-

[9]This finding is in line with a large body of literature studying the "greenium", i.e. the premium difference in Yield-to-Maturity between "comparable" green and conventional bonds. These papers find that evidence of "greenium" in secondary market transaction is absent while some find some evidence on "greenium" in primary market transactions. See, among others, Ehlers and Packer (2017) and Zerbib (2019).

	Total Amount Issued			**Total Amount Outstanding**			**Number issued**		
	Stoxx600	All firms	Fraction Stoxx600	Stoxx600	All firms	Fraction Stoxx600	Stoxx600	All firms	Fraction Stoxx600
Green bonds	99	380	0.26	98	358	0.27	169	1042	0.16
All bonds	2390	4400	0.54	2260	4310	0.52	5476	75898	0.07

Table 1: Descriptive statistics for the amount and number of EUR denominated corporate conventional and green bonds. Data are from *Bloomberg Fixed Income database* and were retrieved in November 1st, 2021. All amounts are in billions of U.S. dollars.

	# distinct issuers	# distinct bonds	# Eligible or Quasi-eligible bonds
EUR denominated conventional bonds issued by Stoxx600	244	2485	1667
EUR denominated green bonds issued by Stoxx600	63	125	114
SEK denominated green bonds (all issuers)	81	282	15

(a) Universe of EUR-denominated bonds issued by Stoxx600 firms and of SEK-denominated corporate green bonds in our sample.

	Mean	Sd	Min	p5	Median	p95	Max
Yield-to-Maturity	0.554	0.879	-0.573	-0.435	0.326	2.475	3.583
Time to Maturity (in years)	6.905	4.456	2.007	2.288	5.748	15.119	44.735
Coupon	1.614	2.155	-0.072	0.000	1.229	4.500	78.000
Amount Issued (in M$)	581.714	630.126	0.457	3.926	550.980	1634.418	6522.120

(b) Characteristics of the EUR-denominated conventional bonds in our sample issued by Stoxx600 firms.

	Mean	Sd	Min	p5	Median	p95	Max
Yield-to-Maturity	0.358	0.575	-0.429	-0.351	0.220	1.462	3.545
Time to Maturity (in years)	8.186	4.060	2.130	2.494	7.685	16.496	21.265
Coupon	1.078	0.763	0.000	0.125	0.900	2.750	3.500
Amount Issued (in M$)	740.160	385.154	28.250	155.470	618.755	1197.620	2848.056

(c) Characteristics of the EUR-denominated green bonds in our sample issued by Stoxx600 firms.

	Mean	Sd	Min	p5	Median	p95	Max
Yield-to-Maturity	1.111	1.241	-0.429	0.115	0.773	3.779	6.336
Time to Maturity (in years)	5.176	4.434	-0.654	2.371	4.638	7.724	63.399
Coupon	1.457	1.381	0.000	0.381	1.090	4.492	8.500
Amount Issued (in M$)	90.678	127.671	1.128	22.119	59.311	182.486	1105.200

(d) Characteristics of the SEK-denominated green bonds in our sample.

Table 2: Descriptive Statistics at the bond level, for those bonds for which we were able to obtain data on Yield-to-Maturity from *DataStream*.

ties" and "Industrials"), with an additional breakdown at BICS Level 2. Financials are among the largest issuers, with particular sub-sector portions by Banks and Real Estate firms. Among the "Utilities" sector, Utility firms rank among the biggest issuers.

We include some further descriptive statistics about green bond issuance in Appendix C.2. Figure C.3 shows the strong growth experienced by corporate green bonds since 2017, with a majority of issuance from Financial and Utility firms. Finally, Figure C.5 depicts the Amount issued in eligible and non-eligible for each BICS Level 1 sector, and we point out the top EUR-denominated corporate green bond issuers in Figure C.6.

5 Empirical Design

5.1 Impact on bond Yield-to-Maturities

In order to evaluate the impact of the ECB green shift on bond prices, we follow the empirical specification Bremus et al. (2021) have designed to study the impact of the CSPP creation on eligible corporate green bonds. We focus instead on the "green shift" announcement on July 8, 2021. The period after this date up to January 1, 2022 corresponds to the *Post* dummy in all of our regressions.

For green bonds, we use the following regression specification:

$$y_{it} = \beta(\text{Eligible} \cdot \text{Post})_{it} + \Gamma_w + \mu_i + \epsilon_{it}, \tag{5.1}$$

where y_{it} is the bid Yield-to-maturity of bond i in day t. We also follow Bremus et al. (2021) by including week fixed effects (denoted by Γ_w), and some bond fixed effects (μ_i) and further controlling for some time-varying factors at the country and at the sector level through some country-by-month as well as sector-by-month fixed effects. Finally, we cluster standard errors at the bond level to robustify against serial correlation in the outcome (Bertrand et al., 2003). ECB-eligible green bonds issued by Stoxx600 corporations represent our treatment group, while our main control group consists in the universe of quasi-eligible (i.e. investment grade) SEK-denominated green bonds. We have also tested (see Appendix D.1) against other control groups, namely non-eligible EUR-denominated green bonds issued by Stoxx600 non-credit corporations, and non-eligible EUR denominated green bonds issued by Stoxx600 credit corporations.

Then, we measure the impact of the ECB "green shift" on ECB-eligible conventional bonds issued by Stoxx600 corporations, depending on the greenness of the issuer. The corresponding regression specification is:

$$y_{it} = \beta(\text{Brown issuer} \cdot \text{Post})_{it} + \Gamma_w + \mu_i + \nu_f + \epsilon_{it}, \tag{5.2}$$

where ν_f are issuer fixed effects. Our treatment group comprises ECB-eligible conventional bonds issued by brown Stoxx600 corporations, while ECB-eligible conventional bonds issued by green Stoxx600 corporations constitute the control group. We identify brown and green issuers according to either a global screening or a sectoral screening, as detailed in the section 4 above.

5.2 Impact on stock market

We study the effect of the ECB green shift on stock market returns using an event study analysis. We compute Cumulative Abnormal Returns (CAR) for the cross sections of Stoxx600 firms following the global three-factor model of Fama and French (1992) to estimate the model expected returns. First, we collect data on European Market Excess Return, HML and SMB factors from French data library available from Kenneth French's website[10]. We estimate the model by regressing each stock return on the global three-factor model which provides the coefficient of interest for each firm f in our sample:

$$R_{ft} = \alpha_f + \beta_{1f} R_m + \beta_{2f} R_{SMB} + \beta_{3f} R_{HML} + \epsilon_{ft}, \tag{5.3}$$

where R_{ft} are the stock return for firm f at day t, R_m is the excess return on the market factor, R_{SMB} is the return on the size factor and R_{HML} is the return on the value factor. We winsorize all three beta estimates at the fifth and ninety-fifth percentiles, as in Becker et al. (2012). The parameters are used to compute the market implied returns and the Abnormal Returns as the difference between the daily realized return and global three-factor model implied returns:

$$\hat{R}_{ft} = \hat{\alpha_f} + \hat{\beta}_{1f} R_m + \hat{\beta}_{2f} R_{SMB} + \hat{\beta}_{3f} R_{HML} \tag{5.4}$$

$$AR_{ft} = R_{ft} - \hat{R}_{ft}. \tag{5.5}$$

Finally, we set a narrow event window of one day before and one day after the event, and sum the Abnormal Returns over the event window to obtain the Cumulative Abnormal Returns for each stock:

$$CAR_f = \sum_j AR_{fj} \mathbb{1}(j \in [-1, +1]) \tag{5.6}$$

5.3 Impact on bond issuance

In order to measure the impact of the ECB announcement on corporate green bond issuance, we study issuer-level weekly changes in cumulative green bond issuance as in the following:

$$y_{ft} = \beta_1 (\text{Treat} \cdot \text{Post})_{ft} + \beta_2 \text{t} + X'_{ft} \eta + \Gamma_w + \nu_f + \epsilon_{ft}, \tag{5.7}$$

where y_{ft} is a vector representing the cumulative number of green bonds or the cumulative amount of green outstanding issued by issuer f in week t, and X_{ft} represents a matrix of issuer-level time-varying characteristics, such as past cumulative green bond issuance or other firm level characteristics.

[10] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

6 Results Price Reaction

In section 6.1 we focus on the specific effects of the ECB announcement on green bonds pricing in the secondary market. Then, in section 6.2 we expand the analysis to conventional bonds pricing, conditional on firms' direct emissions. Finally, in section 6.3, we investigate whether the announcement led to some second order effects on the stocks of Stoxx600 firms.

6.1 Green Bonds

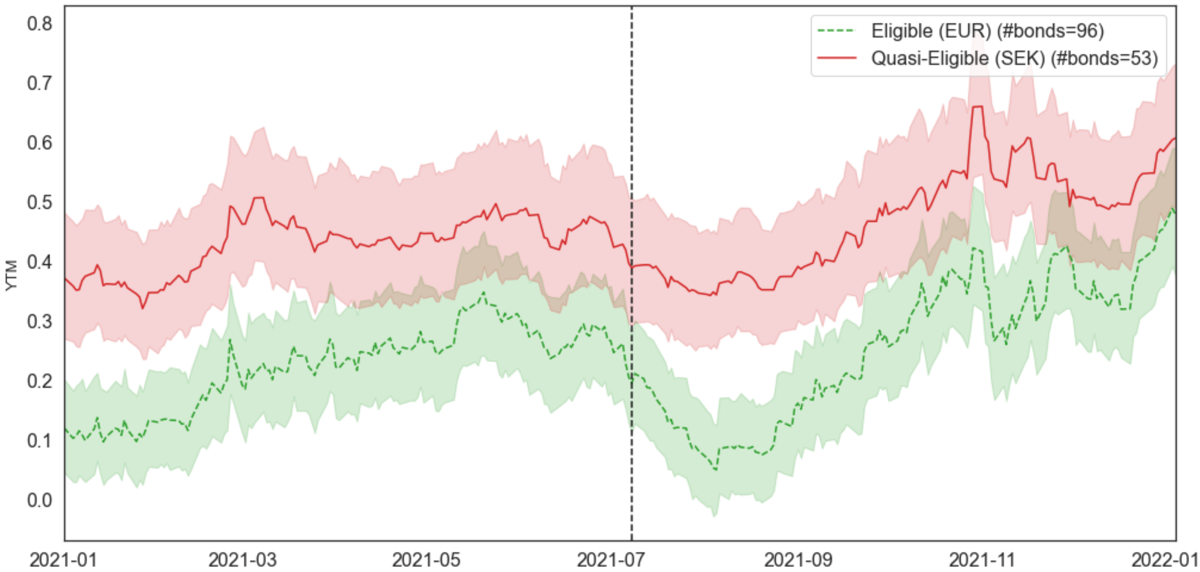


Figure 3: shows the time series of the mean and 95% confidence interval for the Yield-to-Maturity of eligible bonds (Green dashed line) and quasi-eligible green bonds. Quasi-eligible green bonds are investment-grade SEK-denominated green bonds. The vertical dashed line indicates the announcement of the conclusions of the Monetary Policy Strategy Review.

We consider the effects of the ECB announcement on the Yield-to-Maturity of eligible green bonds, which we compare to ineligible corporate green bonds. Our preferred control group consists of quasi-eligible (defined as investment-grade) SEK-denominated green bonds, as it represents bonds of comparable credit worthiness and allows to net out potential re-balancing effects between eligible and non-eligible green bonds within the Euro green bond market, thereby reducing concerns about possible spillover effects[11]. In Appendix D.1, we use an auxiliary control group based on EUR-denominated green bonds that are not investment-grade in order to study the effect of eligibility.

Figure 3 shows the evolution of the average Yield-to-Maturity for the treatment and control groups. As we can see, there was a sharp decrease in the average Yield-to-Maturity of eligible green bonds after the ECB announcement. We formally test this hypothesis in Table 3 and find that, following the ECB announcement, Yield-to-Maturity decreased by an average of 12 bps for eligible green bonds compared to

[11] A potential threat to identification of the effect under investigation are spillovers from green bond market to conventional bond market within the EU as discussed in Bremus et al. (2021) and Pegoraro and Montagna (2022)

quasi-eligible green bonds. The effect is highly statistically significant and robust to considering different fixed effects. It represents about half of the magnitude found by Bremus et al. (2021) in which they study the response to the CSPP announcement in March 2016.

We also investigate whether the effect was subject to some heterogeneity depending on the "greenness" of the bond issuer. In specifications (5) and (6), we focus on subsets of the treatment group based on the global distribution of emissions, and we filter for brown and non-brown issuers. We find that the effect is not significant for brown issuers, while for non-brown issuers the effect is equal to a reduction of 20 bps, and is highly statistically significant. In contrast, in specifications (7) and (8), we focus on subsets of the treatment group based on the sectoral distribution of emissions. We find that the effect is more homogeneous and equal to a reduction ranging between 12 bps and 13 bps. We interpret these findings as indicative that the effects are present for issuers that are not among the worst environmental performers, as they secure a significant 20 bps discount on their green bonds compared to otherwise equivalent SEK-denominated green bonds.

Using our auxiliary control group, we find that Yield-to-Maturity also declined for eligible green bonds, compared to non-eligible EUR-denominated green bonds issued by non-credit Stoxx600 firms (Table D.4). However, the estimated magnitude of the effect is lower (about 5 bps), hinting at some spill-over effects between eligible and non-eligible EUR-denominated green bonds.

Taken together, these findings suggest that the announcement led to a significant reduction in the Yield-to-Maturity of eligible bonds vis-à-vis non eligible ones. The magnitude of the effect is robust against fixed-effect specifications, and amounts to about half of the magnitude found in the case of CSPP and PEPP announcement in Bremus et al. (2021). Some possible explanations to this lower estimated magnitude are that: (i) investors attribute lower importance to the "green shift" announcement, (ii) there is less space for monetary policy manoeuvring on green bonds, which already trade at low Yield-to-Maturities. In addition, we find that the effect is stronger for greener issuers at global and within sector levels. Next, we test whether conventional bond prices were also affected by this ECB announcement.

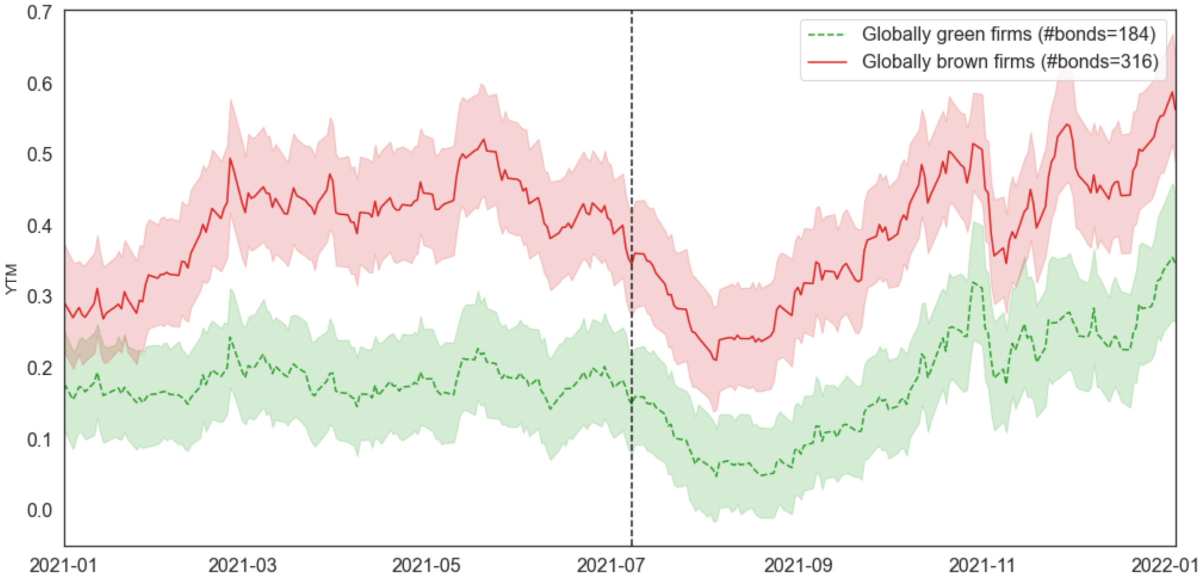
6.2 Conventional Bonds

In this subsection, we focus on the effects on conventional bonds. While the ECB announcement is expected to directly affect green bonds via ECB purchases, conventional bonds could also be affected by a portfolio re-balancing in line with a shift from the principle of 'Market Neutrality' to 'Market Efficiency'. In order to test this hypothesis, we use the two definitions of "greenness" provided in section 4, and we construct two sets of control and treatment groups. The first set is based on the top and bottom quartile of the distribution of emissions in the entire set of Stoxx600 firms, while the second set is based on the top and bottom quartile of the distribution of emissions in each 4-digit sector. First, we consider the relative level emissions in the entire set of Stoxx600 firms and thus test whether investors find credible that the worst performers (e.g. coal companies) would be excluded and "Market neutrality" relaxed. Then, we consider sectoral emissions, more in line with a "Market efficient" tilt towards companies with the lowest emissions in each sector. In both cases, our treatment group is the set of eligible conventional bonds issued by brown firms, whereas our control group comprises eligible conventional bonds issued by green firms. Figures 4a and 4b show the evolution of the average Yield-to-Maturity for the groups based respectively on emissions in the entire and sectoral distribution of emissions.

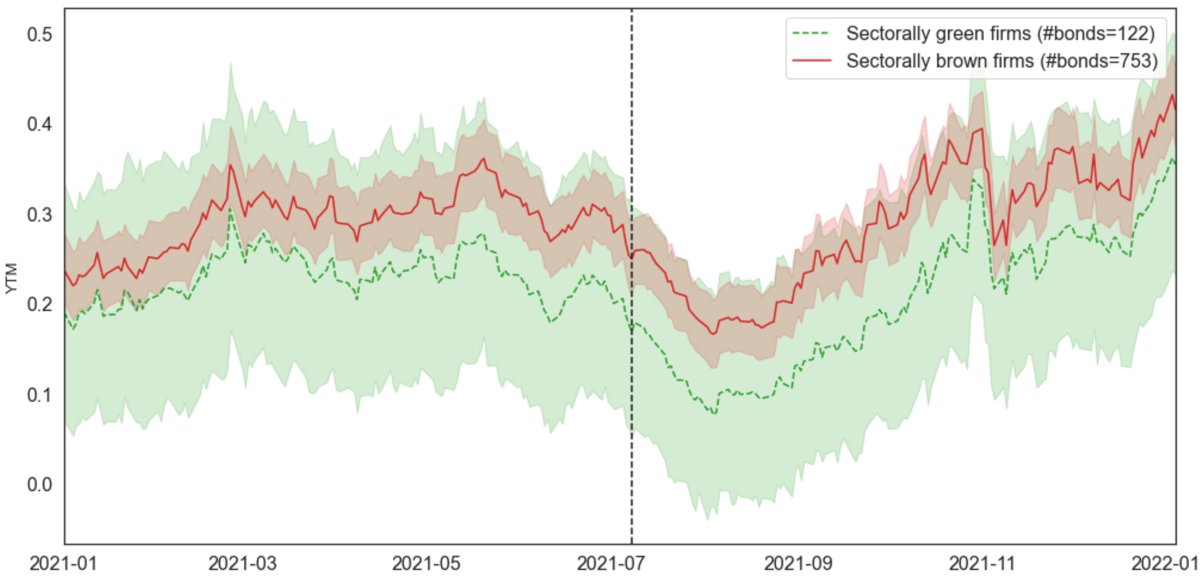
Results for the treatment and control groups based on the distribution of emissions of the entire set of Stoxx600 firms are presented in Columns (1) to (5) of Table 4. We find that, following the ECB announcement, Yield-to-Maturity of eligible conventional bonds issued by brown issuers decreased on average by 14.0 bps to 15.5 bps compared to those of green issuers, depending on the fixed effects included in the regression. More specifically, from specification (1) we find, in line with Figure 4a, that conventional bonds issued by brown firms trade at higher yields compared to those issued by green firms, as we see from the coefficient on the variable *Brown Issuer*, which is positive and highly significant. In addition, we obtain that the difference is significantly reduced following the ECB announcement as highlighted by the estimates of the interaction variable *Brown Issuer* $\times$ *Post* which is negative and highly significant in Columns (2) to (5). We interpret this finding as indicative of investors believing that the ECB will not implement a global screening approach based on emission levels, and will instead maintain its "Market neutrality" stance and keep the most polluting firms in its portfolio.

In column (6) to column (10) of Table 4, in which brown and green issuers are identified within each sector, we find that following the ECB announcement, Yield-to-Maturity declined by 8.9 bps for conventional bonds issued by either green and brown firms. In contrast with the previous specification, we do not find any significant difference between the effects on green and brown firms as highlighted by the non significance of the terms *Brown Issuer* $\times$ *Post* and *Brown Issuer*. This lack of significant reaction at industry level is indicative of investors not believing in the "Market Efficiency" implementation by the ECB in the first place and after the announcement.

The above findings highlight that the ECB announcement has triggered significant market reaction also for conventional bonds, in particular for those issued by the most polluting firms. However, the effect on the Yield-to-Maturity on bonds issued by those firms is negative, and hints at a positive price reaction. We conclude that, with its green shift announcement, the ECB did not manage to push up the price of conventional bond financing for the most polluting firms. Instead, this announcement is more compatible with "Market neutrality" keeping prevalence in investors' view.



(a) Green and brown firms are identified through **global** level direct emissions.

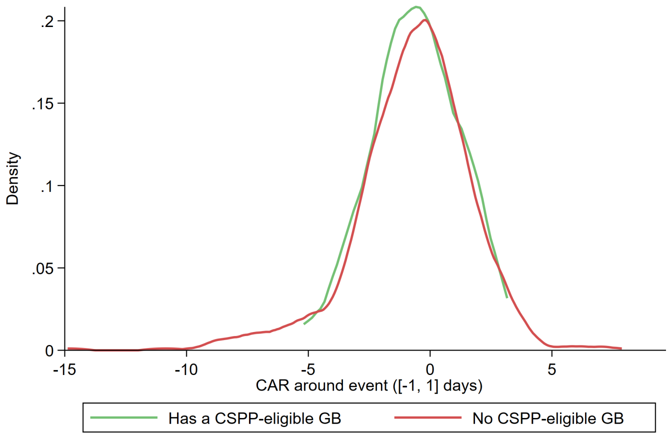


(b) Green and brown firms are identified through **sectoral** level direct emissions.

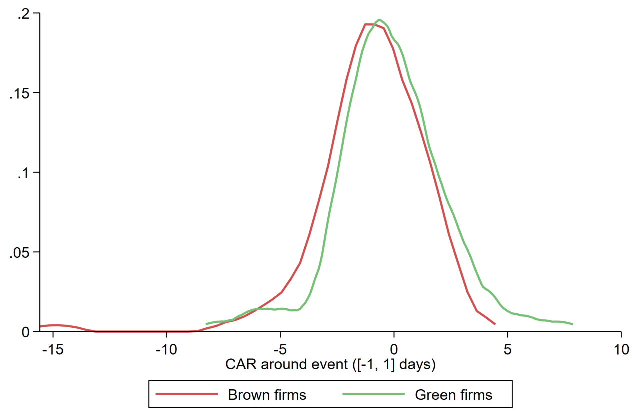
Figure 4: These graphs show the time series of Yield-to-Maturity of eligible bonds for green firms (Green dashed line) and eligible bonds for brown firms.

6.3 Equity

In this subsection, we examine the stock price reaction to assess whether stock markets also responded to the ECB announcement. The event study results are reported using sorted portfolios based on firm characteristics. First, we sort firms depending on whether they have issued an eligible green bond. Kernel densities of the CARs for each group are reported in Panel (a) of Figure 5. The two groups present very similar distributions. In Column (1) of Table 5, we further test whether there is a significant difference in the average and median CAR of the firms sorted on that characteristic. We obtain that both are non significant, suggesting that there was no strong reaction based on a firm having an eligible green bond. When we instead sort firms based on emissions, we obtain that the stock market reaction of green firms is



(a) Depending on whether the firm has an eligible green bond. The green line represents the distribution of the CAR of firms that have an eligible green bond, while the red line is related to those firms that do not have an eligible green bond.



(b) Depending on the greenness of the firm (with respect to global emissions). The green line represents the distribution of green firms, while the red line is related to those brown firms.

Figure 5: Kernel density plot related to the Cumulative Abnormal Returns computed for Stoxx600 firms around the symmetric event window.

positive (Panel (b) of Figure 5). Column (2) of Table 5 shows the average and median CAR when sorting firms based on that characteristic. The positive sign and strong significance for both tests suggests that investors interpret the announcement as being on average more positive for greener firms. We find that a portfolio which is long the bottom quartile of emissions and short the top quartile of emissions earns an average 0.8% (and median reaction of 0.6%) excess return around the announcement date. To benchmark our results, we compare to Flammer (2021)'s result on CAR following green bond issuance. She finds that average CAR in the [-5, 10] window around the event accounts for 0.7% of stock returns.

7 Results Issuance

The previous section shows the statistically and economically significant reaction of EUR-denominated green bonds pricing to the ECB announcement. In this section, we examine whether lower secondary market Yield-to-Maturities for green bonds increases incentives for firms to issue EUR-denominated green bonds, and whether and how corporations increased their green bond issuance after the "green shift" announcement. Figure 6 shows the evolution, at the aggregate level, of the par of newly issued green bonds, and of the number of new green bond issuers. We can see that all segments of the green bond market are growing. We formally test whether, at the issuer level, the ECB announcement accelerated that growth, based on: (i) the domicile of the issuer, (ii) the eligibility status of the issuer.

Based on firm domicile We start by comparing whether firm-week cumulative issuance for firms which are domiciled in the Eurozone or are EUR-denominated green bond issuers react differently compared to issuers that are not domiciled in the Eurozone. Table 6 shows the results of the DiD estimation for

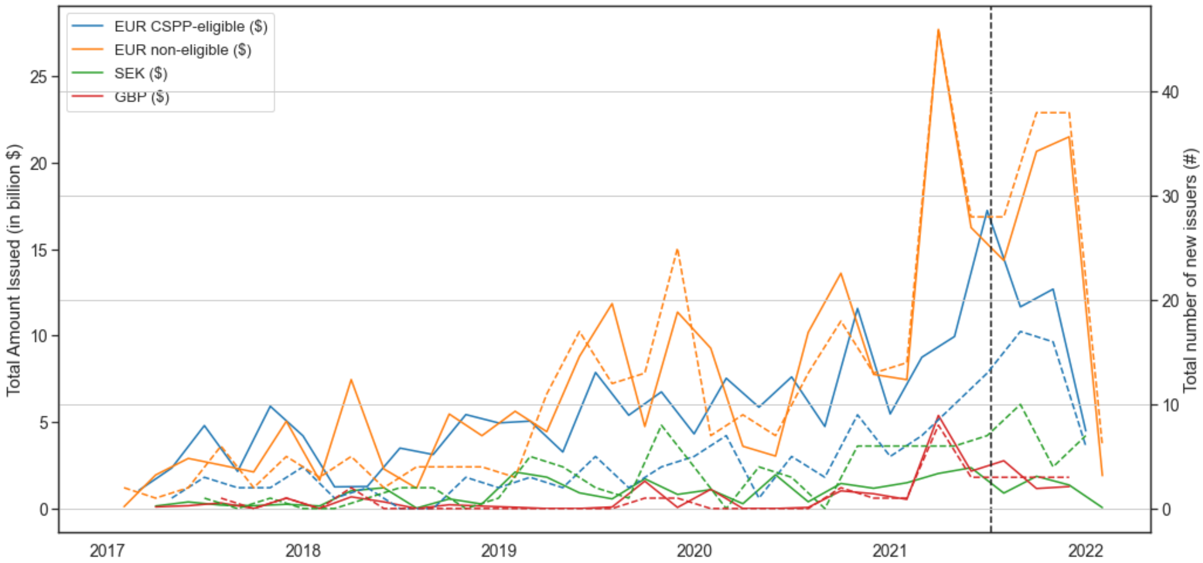


Figure 6: shows the time series of issuance par and the number of new issuers of green bonds for EUR-denominated Eligible green bonds, EUR-denominated not-Eligible green bonds, SEK-denominated green bonds and GBP-denominated green bonds.

the dependent variable of *Cumulative Outstanding Number of Bonds* at firm-week level in Panel (A) and the DiD estimation of the *Cumulative Outstanding Par (in USD billion) of Bonds* at firm-week level in Panel (B). In Column (1) of each panel we test and confirm for the presence of a Time-Trend as seen at aggregate level in Figure 6. The significant trend component confirms the growth of green bonds as an asset class in the Eurozone. In Column (2) of Panel (A) and Panel (B) we estimate the main specification. Post ECB announcement, EUR-denominated green bond cumulative issuance increases on average by 0,176 USD billion. Column (3) and Column (4) of Panel (A) and Panel (B) show that the post treatment effect is even larger for the subset of banks. This is in line with banks being among the largest issuers of green bonds in the Eurozone and with financial intermediaries being able to react fast to changes in economic conditions in debt capital markets. In terms of number of bonds issued, the estimated coefficients in columns (2) and column (3) are not significant while negative and significant in column (4) in the case of non-banks firms.

Within the Eurozone, based on issuer eligibility We next move to the study of differential reaction between eligible and non-eligible EUR-denominated green bonds. Similarly to Todorov (2020) we are interested in understanding whether the ECB-announcement has had catalytic effects for a specific segment of the bond market comprising eligible green bonds. The treatment group in this specification are eligible EUR-denominated green bond issuers whereas the control group are non-eligible EUR-denominated green bond issuers. We remove from the analyses a total of 33 firms which are issuers of both eligible and non-eligible bonds.

Table 7 shows the results of the DiD estimation for the dependent variable of *Cumulative Outstanding Number of Bonds* at firm-week level in Panel (A) and the DiD estimation of the *Cumulative Outstanding*

Par (in USD billion) of Bonds at firm-week level in Panel (B). Specification (2) shows that the interaction term is significant and negative in terms of number of new issued green bonds in Panel (A) and positive and significant for the par of newly issued green bonds in Panel (B). We interpret this finding as indicative of a statically and economically significant increase in eligible green bonds following the ECB-announcement. Interestingly, the effect in terms of par issuance is exacerbated at non-banks level, which shows also an increase in the number of newly issued bonds.

	All firms				Greenness defined over **full economy**		Greenness defined at **sector level**	
	(1)	(2)	(3)	(4)	(5) Brown Issuers	(6) Non-brown Issuers	(7) Brown Issuers	(8) Non-brown Issuers
Eligible $\times$ *Post*	0.124	-0.128***	-0.122***	-0.122***	-0.006	-0.203***	-0.132**	-0.119***
	(1.32)	(-4.56)	(-4.17)	(-4.20)	(-0.31)	(-4.64)	(-2.33)	(-3.47)
Eligible	-0.421***							
	(-2.71)							
Post	-0.143							
	(-1.59)							
Number of distinct bonds	111	111	111	111	47	53	31	69
Adj. R-squared	0.077	0.844	0.837	0.836	0.860	0.833	0.748	0.857
Bond FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country x Month FE	No	No	Yes	No	No	No	No	No
Sector x Month FE	No	No	No	Yes	Yes	Yes	Yes	Yes

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 3: Effect of the ECB green shift announcement on the Yield-to-Maturity of eligible green bonds, compared to quasi-eligible (investment-grade, SEK-denominated) green bonds.

	Greenness defined over **full economy**					Greenness defined at **sector level**				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post x *Brown Issuer*	-0.024*	-0.029***	-0.029***	-0.011	-0.015*	0.028*	0.012	0.012	0.019	0.005
	(-1.66)	(-2.81)	(-2.81)	(-0.62)	(-1.96)	(1.96)	(1.14)	(1.14)	(1.58)	(0.84)
Brown Issuer	0.225***					0.064				
	(4.37)					(0.90)				
Post	-0.081***					-0.097***				
	(-7.42)					(-7.51)				
Number of distinct bonds	487	487	487	487	487	842	842	842	842	842
Adj. R-squared	0.034	0.967	0.967	0.969	0.974	0.006	0.963	0.963	0.966	0.970
Bond FE	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Week FE	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Issuer FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Country x Month FE	No	No	No	Yes	No	No	No	No	Yes	No
Sector x Month FE	No	No	No	No	Yes	No	No	No	No	Yes

t statistics in parentheses

* $p<0.10$, ** $p<0.05$, *** $p<0.01$

Table 4: Effect of the ECB green shift announcement on the Yield-to-Maturity of eligible conventional bonds.

	Firms with a CSPP eligible green bond vs firms with no CSPP eligible green bond	Green firms vs brown firms (wrt global scope 1 emissions)
Mean CAR	-0.105 (0.776)	0.817*** (0.009)
Observations	658	236

p-values in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Table 5: T-test on the difference of the mean and median (Wilcoxon test) of portfolios formed on the criteria described in the table header. Units are percentage points. p-values are in parenthesis.

Panel A: Number of bonds

	All issuers		Banks	Non-Banks
	(1)	(2)	(3)	(4)
Post	0.582***	0.058	-0.248	0.458***
	(3.93)	(0.22)	(-0.80)	(3.41)
Time trend	0.049***	0.007	-0.011***	0.020***
	(6.03)	(0.97)	(-3.82)	(12.28)
Post × *Treated*		0.059	-0.260	-0.257*
		(0.27)	(-0.83)	(-1.76)
Lagged Number of Bonds Issued		1.636***	1.942***	0.922***
		(4.39)	(160.00)	(11.87)
Lagged Amount Issued		-0.776***	1.631***	-0.451***
		(-3.00)	(16.54)	(-3.35)
Observations (Issuer-Week)	88308	81972	15642	66330
Adj. R-squared	0.550	0.828	0.844	0.923
Week FE	No	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes

Panel B: Par of bonds (in billion $)

	All issuers		Banks	Non-Banks
	(1)	(2)	(3)	(4)
Post	0.163***	-0.020	-0.076**	0.008
	(8.26)	(-0.60)	(-2.47)	(0.22)
Time trend	0.015***	0.008***	0.010***	0.008***
	(12.66)	(10.12)	(33.84)	(9.39)
Post × *Treated*		0.186***	0.306***	0.133***
		(4.40)	(9.72)	(2.92)
Lagged Number of Bonds Issued		0.015	0.034***	-0.017
		(1.11)	(28.30)	(-0.97)
Lagged Amount Issued		0.652***	0.620***	0.690***
		(10.40)	(63.09)	(9.06)
Observations (Issuer-Week)	88308	81972	15642	66330
Adj. R-squared	0.836	0.899	0.831	0.912
Week FE	No	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes

Note: T-statistics are in parentheses. Significance levels are indicated by * $<$ .1, ** $<$ .05, *** $<$ .01. Standard errors are clustered at the issuer level.

Table 6: Panel (A) shows the estimates from specification 5.7 of regressing cumulative number of green bonds issued at week-issuer level on a set of control variables and "PostxTreatment" effect. Similarly, Panel (B) estimates the specification 5.7 using as dependent variable cumulative amount of green bond issued at week-issuer level on a set of control variables and "PostxTreatment" effect. The treatment sample is the set of issuers which have issued EUR-denominated green bonds or are incorporated in the eurozone before the treatment under consideration whereas the control group is the set of issuers which have issued only non-eligible green bonds. Column (3) and Column (4) estimate specification 5.7 on the subset of firms which are classified as "Banks" and as "Non-Banks" corporate issuers based on BICS Level 2 classification.

Panel A: Number of bonds

	All issuers		Banks	Non-Banks
	(1)	(2)	(3)	(4)
Post	0.582***	-0.143**	-0.379*	0.143***
	(14.83)	(-2.43)	(-1.93)	(7.29)
Time trend	0.049***	0.017***	0.002	0.022***
	(51.94)	(11.37)	(0.40)	(44.16)
Post × Treated		-0.706***	-3.828***	0.576***
		(-3.04)	(-6.40)	(6.20)
Lagged Number of Bonds Issued		2.226***	2.117***	1.018***
		(272.13)	(125.26)	(108.36)
Lagged Amount Issued		-1.476***	1.287***	-0.481***
		(-45.78)	(8.96)	(-33.17)
Observations (Issuer-Week)	88308	30690	8514	22176
Adj. R-squared	0.550	0.857	0.865	0.918
Week FE	No	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes

Panel B: Par of bonds (in billion $)

	All issuers		Banks	Non-Banks
	(1)	(2)	(3)	(4)
Post	0.163***	0.144***	0.143***	0.156***
	(31.71)	(14.62)	(8.42)	(13.05)
Time trend	0.015***	0.012***	0.010***	0.013***
	(120.69)	(48.68)	(23.72)	(44.12)
Post × Treated		0.285***	0.169***	0.449***
		(7.31)	(3.28)	(7.93)
Lagged Number of Bonds Issued		0.032***	0.035***	-0.075***
		(23.20)	(23.84)	(-13.19)
Lagged Amount Issued		0.641***	0.696***	0.757***
		(118.24)	(56.15)	(85.58)
Observations (Issuer-Week)	88308	30690	8514	22176
Adj. R-squared	0.836	0.921	0.876	0.930
Week FE	No	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes

Note: T-statistics are in parentheses. Significance levels are indicated by * < .1, ** < .05, *** < .01. Standard errors are clustered at the issuer level.

Table 7: Panel (A) shows the estimates from specification 5.7 of regressing cumulative number of green bonds issued at week-issuer level on a set of control variables and "PostxTreatment" effect. Similarly, Panel (B) estimates the specification 5.7 using as dependent variable cumulative amount of green bond issued at week-issuer level on a set of control variables and "PostxTreatment" effect. The treatment sample is the set of issuers which have issued only ECB eligible green bonds whereas the control group is the set of issuers which have issued only non-eligible green bonds. Column (3) and Column (4) estimate specification 5.7 on the subset of firms which are classified as "Banks" and as "Non-Banks" corporate issuers based on BICS Level 2 classification.

8 Conclusions

This paper sheds light on the role of central banks in fostering the adoption of green financing instruments. Our study focuses on the recent announcement by the European Central Bank of its *Monetary Policy Strategy Review* in July 2021. The review, which had the objective of recalibrating the ECB monetary policy objectives to changes to the economic conjuncture, unexpectedly highlighted the role of climate change on the transmission of monetary policy. It led the ECB to propose a road-map to incorporate climate considerations within its monetary policy operations, while remaining within the boundaries of its mandate.

In line with a large body of literature studying the effects of ECB asset purchase programmes (De Santis and Zaghini (2019), Bremus et al. (2021), Todorov (2020), Grosse-Rueschkamp et al. (2019), Koijen et al. (2021)) we find strong evidence of the role of the ECB in the Eurozone green bond market. We find that: (i) EUR-denominated green bonds reacted with a statistically and economically significant reduction in average YTM when compared to non-Eurozone quasi-eligible green bonds, (ii) the ECB announcement did not lead to a decrease in the cost of conventional bond finance for green firms, (iii) on the contrary, green firms' stock reacted positively to the ECB announcement.

Motivated by our first finding, we study whether the decrease in the cost of green bond finance led to a change in green bond issuance behaviour of firms in the Eurozone. We compare the response of issuers domiciled in the Eurozone with other jurisdictions and find that Eurozone green bond issuers substantially increased green bond issuance following the ECB announcement, with the effect being more pronounced for banks. Then, we study the effect of the ECB announcement within the Eurozone by comparing the issuance behaviour between ECB-eligible vis-à-vis ECB-ineligible green bonds. We find that issuance of ECB-eligible green bonds increased in the Eurozone compared to non-eligible green bonds, with the effect being stronger within the non-banking sector.

Overall, our findings provide evidence of the positive effect of including green considerations within the ECB monetary policy operations. They are important for central banks when assessing the potential impact of their commitment to supporting green financing instrument within their monetary policy operations.

References

J. Azar, M. Duro, I. Kadach, and G. Ormazabal. The Big Three and Corporate Carbon Emissions Around the World. page 78, Dec. 2020.

M. Baker, D. Bergstresser, G. Serafeim, and J. Wurgler. Financing the Response to Climate Change:

The Pricing and Ownership of U.S. Green Bonds. Working Paper 25194, National Bureau of Economic Research, Oct. 2018. Series: Working Paper Series.

B. Becker, D. Bergstresser, and G. Subramanian. Does Shareholder Proxy Access Improve Firm Value? Evidence from the Business Roundtable Challenge. Working Paper 17797, National Bureau of Economic Research, Feb. 2012. Series: Working Paper Series.

M. Bertrand, E. Duflo, and S. Mullainathan. How much should we trust differences-in-differences estimates? page 32, June 2003.

F. Bremus, F. Schuetze, and A. Zaklan. The Impact of ECB Corporate Sector Purchases on European Green Bonds. *SSRN Electronic Journal*, 2021. ISSN 1556-5068.

Z. Chen, A. A. Lookman, N. Schürhoff, and D. J. Seppi. Rating-Based Investment Practices and Bond Market Segmentation. *Review of Asset Pricing Studies*, 4(2):162–205, Dec. 2014.

T. Chordia, A. Goyal, Y. Nozawa, A. Subrahmanyam, and Q. Tong. Are Capital Market Anomalies Common to Equity and Corporate Bond Markets? An Empirical Investigation. *Journal of Financial and Quantitative Analysis*, 52(4):1301–1342, Aug. 2017.

R. A. De Santis and A. Zaghini. *Unconventional monetary policy and corporate bond issuance.* Publications Office, LU, Nov. 2019.

P. Deschryver and F. de Mariz. What Future for the Green Bond Market? How Can Policymakers, Companies, and Investors Unlock the Potential of the Green Bond Market?, 2020.

S. Dikau and U. Volz. Central bank mandates, sustainability objectives and the promotion of green finance | Elsevier Enhanced Reader, 2021.

T. Ehlers and F. Packer. Green bond finance and certification. Sept. 2017.

R. Fahlenbrach and E. Jondeau. Greening the Swiss National Bank's Portfolio. *SSRN Electronic Journal*, 2021. ISSN 1556-5068. doi: 10.2139/ssrn.3906654.

E. F. Fama and K. R. French. The Cross-Section of Expected Stock Returns, 1992.

E. F. Fama and K. R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, Feb. 1993. doi: 10.1016/0304-405X(93)90023-5.

C. Flammer. Corporate green bonds. *Journal of Financial Economics*, 142(2):499–516, Nov. 2021.

R. Gibson Brandon, P. Krueger, and S. F. Mitali. The Sustainability Footprint of Institutional Investors: ESG Driven Price Pressure and Performance. page 77, Jan. 2021.

B. Grosse-Rueschkamp, S. Steffen, and D. Streitz. A capital structure channel of monetary policy. *Journal of Financial Economics*, 133(2):357–378, Aug. 2019.

L. P. Hansen. Central banking challenges posed by uncertain climate change and natural disasters, Sept. 2021.

R. S. Koijen, F. Koulischer, B. Nguyen, and M. Yogo. Inspecting the mechanism of quantitative easing in the euro area | Elsevier Enhanced Reader, 2021.

L. Liebich, L. Nöh, F. Rutkowski, and M. Schwarz. Current developments in green finance. Working Paper 05/2020, Arbeitspapier, 2020.

M. Papoutsi, M. Piazzesi, and M. Schneider. How unconventional is green monetary policy? page 55, Oct. 2021.

L. Pastor, R. F. Stambaugh, and L. A. Taylor. Sustainable Investing in Equilibrium. page 54, June 2020.

S. Pegoraro and M. Montagna. ISSUANCE AND VALUATION OF CORPORATE BONDS WITH QUANTITATIVE EASING. page 61, 2022.

L. Reichlin, K. Adam, W. J. McKibbin, M. McMahon, R. Reis, G. Ricco, and B. Weder di Mauro. The ECB strategy: The 2021 review and its future, 2021.

K. Todorov. Quantify the quantitative easing: Impact on bonds and corporate debt issuance. *Journal of Financial Economics*, 135(2):340–358, Feb. 2020.

A. Zaghini. The CSPP at work: Yield heterogeneity and the portfolio rebalancing channel | Elsevier Enhanced Reader, Dec. 2017.

O. D. Zerbib. Is There a Green Bond Premium? The yield differential between green and conventional bonds. page 67, 2019.

A Additional Institutional Details

Green Bonds in CSPP

As part of the CSPP program, the ECB has purchased eurozone green bonds[12]. In the 2018 ECB Economic Bulletin, ECB economists look at the composition of eurosystem Green Bonds holdings and discuss the extent to which APP purchases have affected the adoption of green bonds.

In its original implementation, the CSPP purchases follow the principle of "market neutrality", by which net purchases are guided by the proportion of market value of all eligible bonds by economic sector and rating groups. By doing so, the portfolio allocation does not include any screening based on environmental or other criteria.

[12]In 2018 the ECB published an economic bulletin on the green bonds purchased in the CSPP. See:https://www.ecb.europa.eu/pub/economic-bulletin/focus/2018/html/ecb.ebbox201807_01.en.html

Between 2013 and 2018, total euro-denominated green investment grade issuance accounted for 24% of total global net green issuance. In mid 2017, total euro-denominated net issuance's surpassed for the first time usd-denominated issuance's and accounted for approximately USD 60 billion. In perspective to the total issuance of bonds in the eurozone, euro-denominated green investment grades accounted for approximately 1% of total euro-denominated debt supply (ECB, 2018).

In the CSPP-eligible universe, green bonds account for 4% of the eligible universe. Interestingly, in terms of industry distribution, green bond issuance are more concentrated in carbon-intensive sectors such as utilities, infrastructure and real estate and transportation. These sectors taken together account for 35% of CSPP-eligible universe but 94% of CSPP-eligible green bond issuance (ECB,2018).

The ECB provides a weekly list of ISINs included in its CSPP portfolio[13]. Figure xxx shows the cumulative number of green bond ISINs included in the CSPP portfolio since its inception.

Sovereign and supranational green bond issuances have also been purchased as part of the PSPP but account for less than 1% of PSPP-eligible universe as of 2018. Multilateral development banks such as the European Investment Bank and agencies like Kreditanstalt für Wiederaufbau have been issuing green bonds since early 2000 while governments' issuances have been increasing since 2020 and the first eurozone issuance date to 2017 with the French Treasury. As of 2018, the eurosystem hold 24% of outstanding sovereign and supranational euro-denominated green bonds issuance which is in line with the total PSPP holdings compared to the PSPP-eligible universe.

A.1 CSPP Institutional Details

The Corporate Sector Purchase Program ("CSPP") represents one of the Asset Purchase Programs ("APP") initiated by the European Central Bank ("ECB") in the aftermath of the great financial crisis. The CSPP was initiated in 2016 and targeted investment grade euro-denominated corporate bonds from issuers located in the euro area. Among the APP, the ABS purchase program (ABSPP), the Covered Bond purchase program (CBPP) and the Public Sector Purchase Program (PSPP) were aimed at purchasing eurozone sovereign and collateralized securities in the attempt of alleviating financing costs for eurozone firms and increase real investments. Despite the positive effects on economic growth and financing conditions achieved in the eurozone (Memertzis and Wolff 2016), these measures were not able to restore inflation rate on its below (but close) 2% target in the medium term[14].

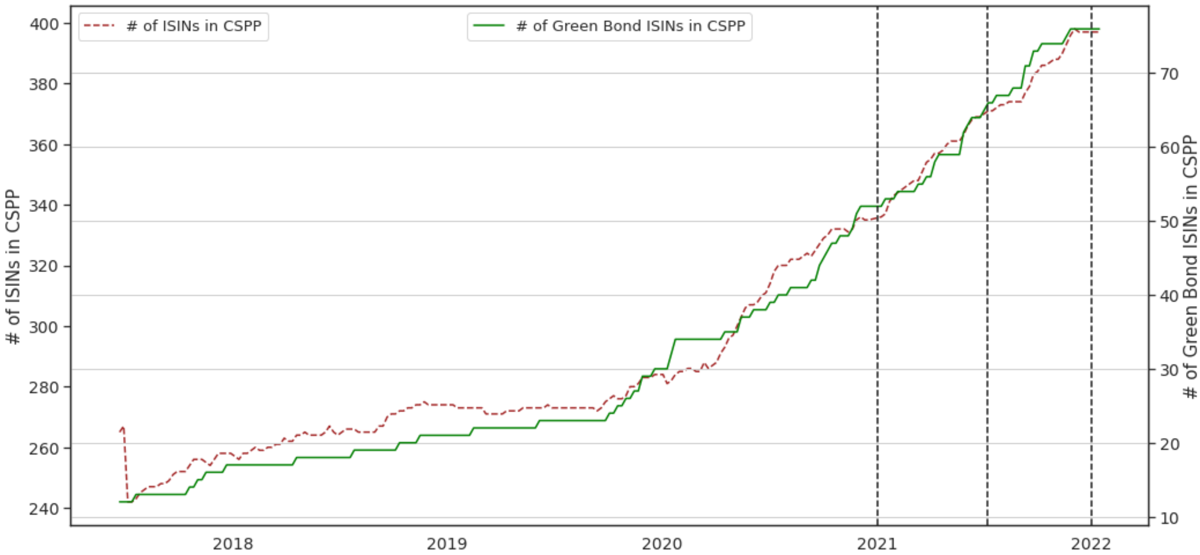
Between October 2014 and December 2018 monthly net purchases under the APP represented a sizable portion of the eurozone capital market and representing a substantial portion of EU GDP countries[15], with the PSPP representing the most sizable portion, followed by the CBPP and CSPP programs:

- EUR 60 billion from March 2015 to March 2016

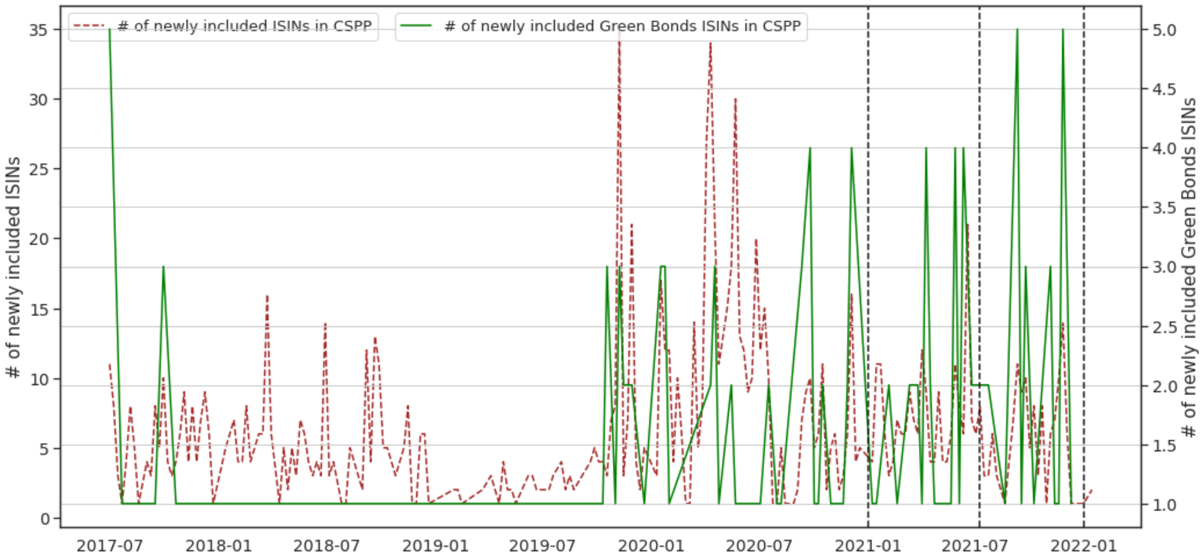
[13] See: https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html#cspp

[14] In line with the ECB primary objective of price stability, inflation rate target is below, but close to, 2% over the medium term.

[15] According to Todorov (2018), the APP accounted for approximately 40% of Eu GDP as of Max 2017.



(a) This graph shows the time series of the number of green bonds included in the CSPP (green line, RHS) and the number of ISINs included in the CSPP program (red dashed line, LHS) between 2017 and 2022. Source: ECB data.

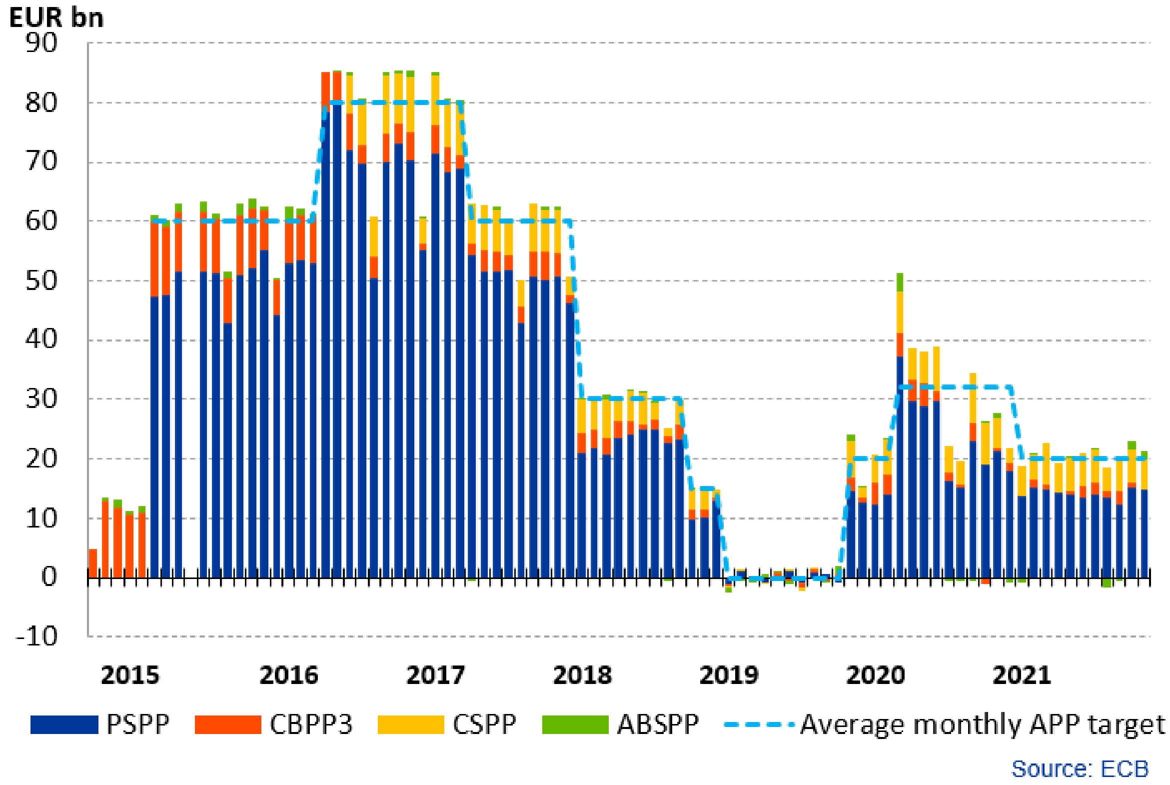


(b) This graph shows the time series of the number of newly included green bonds in the CSPP (green line, RHS) and the number of newly included ISINs in the CSPP program (red dashed line, LHS) between 2017 and 2022. Source: ECB data.

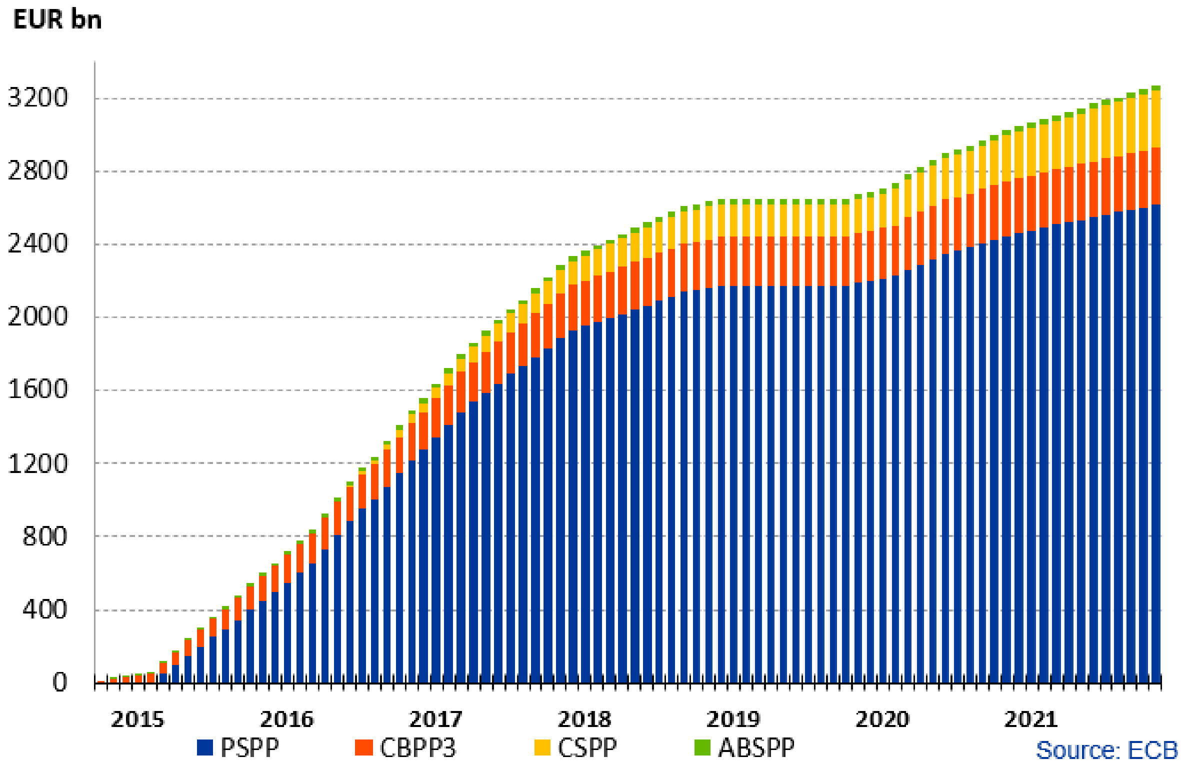
- EUR 80 billion from April 2016 to March 2017
- EUR 60 billion from April 2017 to December 2017
- EUR 30 billion from January 2018 to September 2018
- EUR 15 billion from October 2018 to December 2018

From January 2019 to October 2019 the ECB started reinvesting the maturing securities while the APP were restarted in November 2019 at a EUR 20 billion monthly pace.

The CSPP targeted non-bank investment grade non-secured debt in the eurozone. As of December 2021, EUR 308 billion are held in the CSPP portfolio. These amounts include also securities purchased as part of the Pandemic Emergency Purchase Program (PEPP), which was started in March 2020 in response to



(a) This graph shows the time series of net APP purchases by the eurosystem between 2015 and 2021.



(b) This graph shows the time series of cumulative APP purchases by the eurosystem between 2015 and 2021.

the COVD-19 shock. The PEPP included purchases of private and public sector securities for a total cap of EUR 1850 billion. As of December 2021, a cumulative of EUR 1565 billion were purchased under the PEPP umbrella, with PSPP and CSPP accounting to 91.1% and 2.5% respectively. The first review of the PEPP program is expected in March 2022 but the program will be terminated by the ECB Governing

Council subject to end of the COVID-19 crisis for the eurozone.

The announcement of the CSPP program unfolded in two separated phases. On 10 March 2016, ECB officials announced the expansion of the APP programs with the inclusion of the CSPP program without further specification on the exact details of the pace and eligible securities. On 21 April 2016, in a second announcement, the ECB published official eligibility criteria for the CSPP.

The patterns of communication followed by the ECB is similar in the case of the inclusion of sustainability criteria within its CSPP program. On the 6th of July 2021, the ECB announced the conclusion of its Monetary Policy Review, in which climate change consideration played an primary role. Among possible measures, the ECB has curbed a special role to the CSPP and has planned to provide throughout 2022 further details on the exact implementation timeline and details.

B Institutional Details on Green Bonds

Green Bond Label and Certification process

Green Bonds are debt financial instruments directed at financing firms' green projects/assets. Differently from conventional bonds, "Green Bonds" have a "Use-of-Proceeds" structure in which investors capital are directly channelled into the stated projects/assets in the green bond prospectus.Generally, funds are "earmarked" compared to general corporate funds in order to provide transparency to green bonds investors that funds are solely directed towards the intended target.

The financial industry has developed ad-hoc Green Bond frameworks and taxonomies for issuers and investors for what defines "Green" bonds and "Green" assets/projects with the objective to increase standardization and transparency in the market. At their discretion, issuers can reference in their bond documentation that the bond is a "Green Bond" and if the "Green" label is referenced to a particular Green Bond framework or principle and provide investors with third party assurance and verification, i.e. the practise of providing third party verification of compliance of the bond prospectus with the referred framework or taxonomy[16].

Multiple market standards are to date available for issuers in several major jurisdictions. The ICMA Green Bonds Principles ("GBPs") is among the first and most widely referenced by Green Bonds issuers. The GBPs define a set of principles to guide issuers in labelling Green Bonds and eligibility criteria for underlying projects/assets to be considered material. Green Bonds which comply with the GBP need to satisfy four core principles:

1. *Use of Proceeds:* GBP eligible Green projects/assets need to provide environmental benefits. The GBP considers only broad eligible environmental categories to which projects need to adhere: climate change mitigation, climate change adaptation, natural resource conservation, biodiversity

[16] See Ehlers et al. (2017) for further information. Report accessible at the following link: https://www.bis.org/publ/qtrpdf/r_qt1709h.htm

conservation, and pollution prevention and control;

2. *Process for Project Evaluation and Selection:* issuers need to clearly communicate to investors in relation to: (i) the environmental objective targeted by the green bond, (ii) evaluation process of environmental sustainability by the issuer, (iii) complementary information regarding the use of proceeds including social and environmental risks;

3. *Management of Proceeds:* related to the operation details of the separation of funds compared to general corporate funds;

4. *Reporting:* issuers need to provide updated reporting on the use of proceeds and update them at least annually.

Green Bonds Label and Credit Rating

Green Bond ratings and labels need not to be confused with opinion on an issuer environmental risk exposure.

An important point in relation to green bonds is whether these instruments provide themselves a hedge against environmental risks for investors. These environmental related risks are categorized by the Task Force on Climate-related Financial Disclosures ("TCFD") in Transition Risks, i.e. risks stemming from policy/regulatory or technology shocks, and Physical Risks, i.e. risks originating from natural disasters and change in climate patterns such as global warming. To the extent that green bond issuers are less exposed to environmentally related risks, investing in green bonds could provide a hedge against these shocks. However, green bonds per-se provide a rather limited risk management device in that respect. On the one hand, majority of green bonds are claims to the overall issuers' operations and the green label per-se does not provide an opinion to an issuer exposure to those risks. Furthermore, by the fact that the majority of green bond corporate issuers are in sectors such as energy and industrials, which are notable more exposed to environmental credit risk, investing in green bonds from issuers in these sectors potentially provides considerable environmental risk exposure, Ehlers et al. (2017).On the other hand, issuance of green bonds signals an issuer's commitment to reducing its climate risks exposure which is in line with the finding of Flammer (2020) of investor's positive reaction to firm's announcement of green bond issuance.

C Additional Data Details

C.1 Firm-level descriptive statistics

Variable name	Description	Source	Label
Green bond issuer	A firm is defined as a green bond issuer if it has a strictly positive amount issued in green bonds.	Bloomberg	
Yield to maturity	The yield of a bond calculated until maturity.	Datastream	$y_{i,t}$
Is CSPP	Equal to 1 for firms which have issued at least one eligible debt security which has been or is currently within the CSPP portfolio of the ECB.	ECB	
GICS code	Global Industry Classification Standard industry taxonomy. Sectors are defined using GICS industry classification with 4-digits code. In particular, inside the "Finance" sector (GICS codes from 4010 to 4030), we distinguish **non-credit institutions (GICS codes from 4020 to 4030)** such as Insurance, Real Estate companies, etc. from **credit institutions (GICS codes from 4010 to 4019)**. The latter are not CSPP eligible.	Compustat	
CSPP eligible	Equal to 1 for CSPP-eligible bonds.	Bloomberg	

Table C.1: Variable definitions and sources.

	Mean	Sd	Min	p5	Median	p95	Max	Obs.
Size	9.669	1.835	4.586	6.880	9.465	13.026	15.140	596
Leverage	0.270	0.162	0.000	0.022	0.258	0.542	0.968	579
Sales	2.0e+04	4.1e+04	23.819	481.800	6247.000	8.3e+04	4.3e+05	453
Credit Institutions	0.059	0.235	0.000	0.000	0.000	1.000	1.000	596
Scope 1 Emissions	2.9e+06	1.3e+07	0.000	142.200	3.2e+04	1.6e+07	1.7e+08	489
Scope 2 Emissions	4.7e+05	1.4e+06	0.000	503.000	5.4e+04	2.4e+06	1.3e+07	486
Scope 1 Intensity	177.634	606.691	0.000	0.228	10.904	930.933	6986.667	367
Scope 2 Intensity	38.757	74.791	0.000	0.518	12.986	190.812	581.029	365
Amount EUR conventional bonds issued (in M$)	3385.856	1.2e+04	0.000	0.000	0.000	1.5e+04	1.6e+05	596
Amount EUR green bonds issued (in M$)	191.427	886.342	0.000	0.000	0.000	1188.230	1.5e+04	596
# EUR conventional bonds issued	6.930	30.229	0.000	0.000	0.000	19.000	466.000	596
# EUR green bonds issued	0.309	1.423	0.000	0.000	0.000	2.000	20.000	596

Table C.2: Descriptive Statistics at the issuer level, for the Stoxx 600 firms. *Size* is the natural logarithm of the book value of total assets (in US dollars). *Leverage* is defined as the ratio of debt (as defined by the addition of long term debt and debt in current liabilities) to the book value of total assets. *Intensities* are computed using the ratio of emissions to *Sales*.

	Size	Leverage	# EUR green bonds issued	# EUR conventional bonds issued	Scope 1 Emissions	Scope 1 Intensity	Green f
Size	1						
Leverage	-0.0374	1					
# EUR green bonds issued	0.259***	0.00831	1				
# EUR conventional bonds issued	0.337***	-0.0305	0.345***	1			
Scope 1 Emissions	0.147***	-0.0351	0.0922**	-0.0238	1		
Scope 1 Intensity	0.130**	-0.0314	0.160***	0.0183	0.718***	1	
Green firm (global scope 1 emissions)	-0.0577	-0.0190	-0.0495	0.00901	-0.130***	-0.122**	
Brown firm (global scope 1 emissions)	0.173***	0.0673	0.0653	-0.0415	0.382***	0.389***	
Green firm (sectoral scope 1 emissions)	-0.167***	0.0263	-0.0504	-0.0317	-0.135***	-0.163***	
Brown firm (sectoral scope 1 emissions)	0.326***	0.0431	0.108***	0.151***	0.304***	0.285***	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.3: Correlation table between the main variables used for emission levels and intensities at different GICS classification granularity(4d: four digits, 2d: two digits, full: full digits) and firm level variables related to green bond issuance and size. Lower rank means lower emissions.

C.2 Corporate green bonds

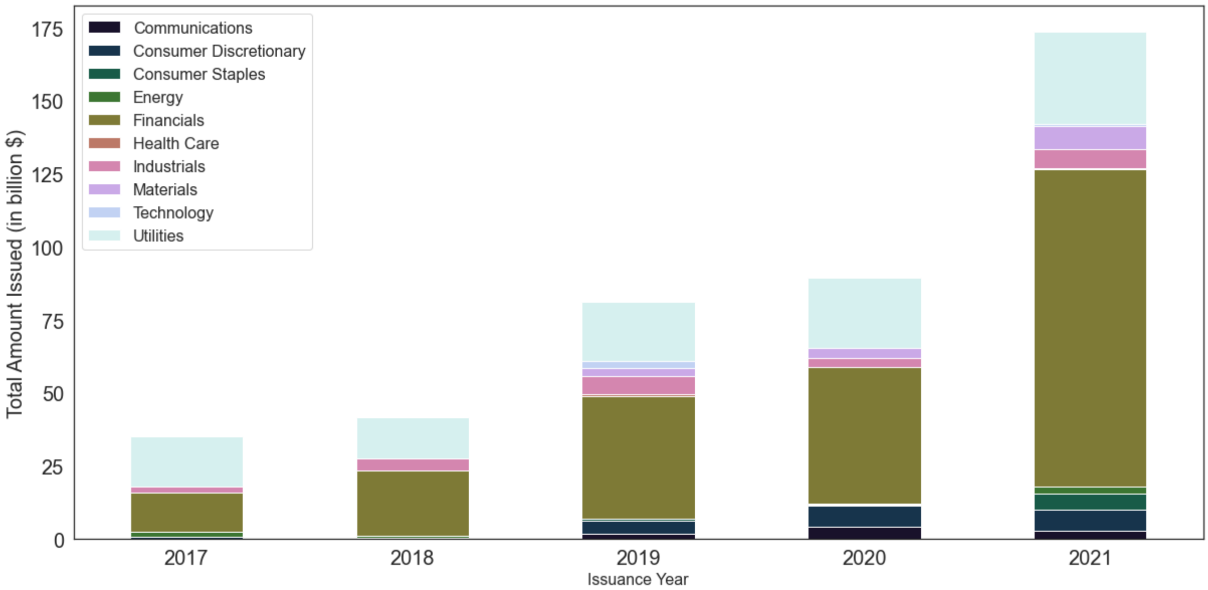


Figure C.3: shows the total amount of EUR-denominated issuance of green bonds (in USD billion) by year between 2017 and 2021 by BICS Level 1.

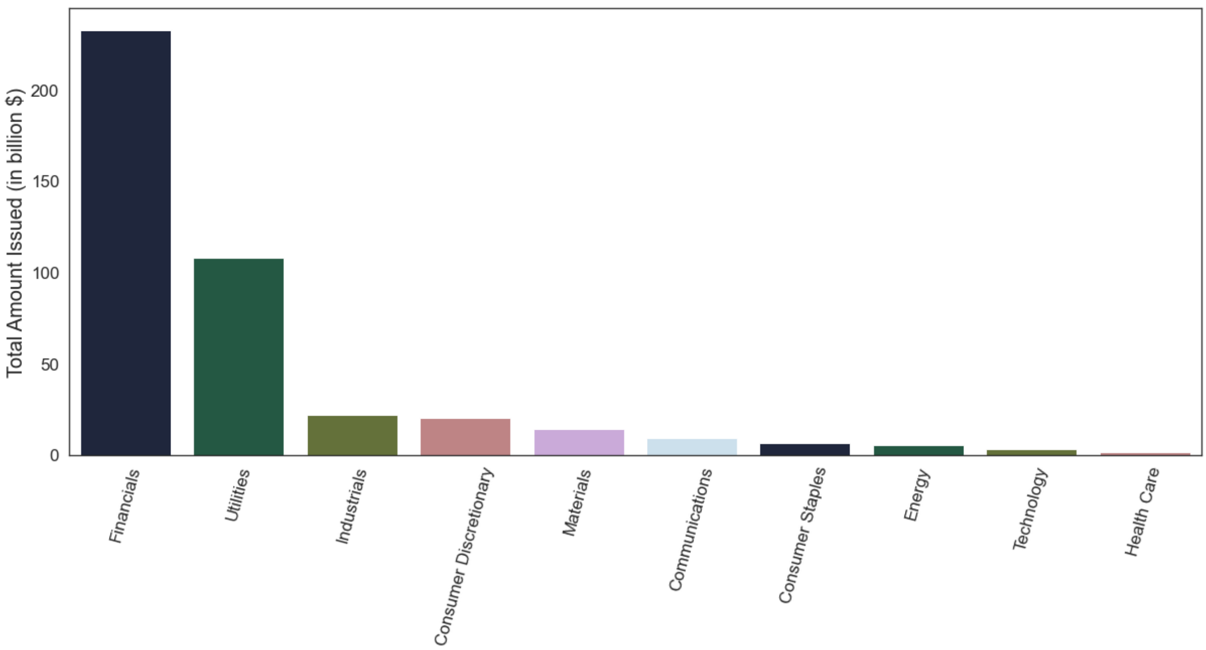


Figure C.4: shows the aggregate amount of EUR-denominated issuance of green bonds (in USD billion) between 2017 and 2021 by BICS Level 1.

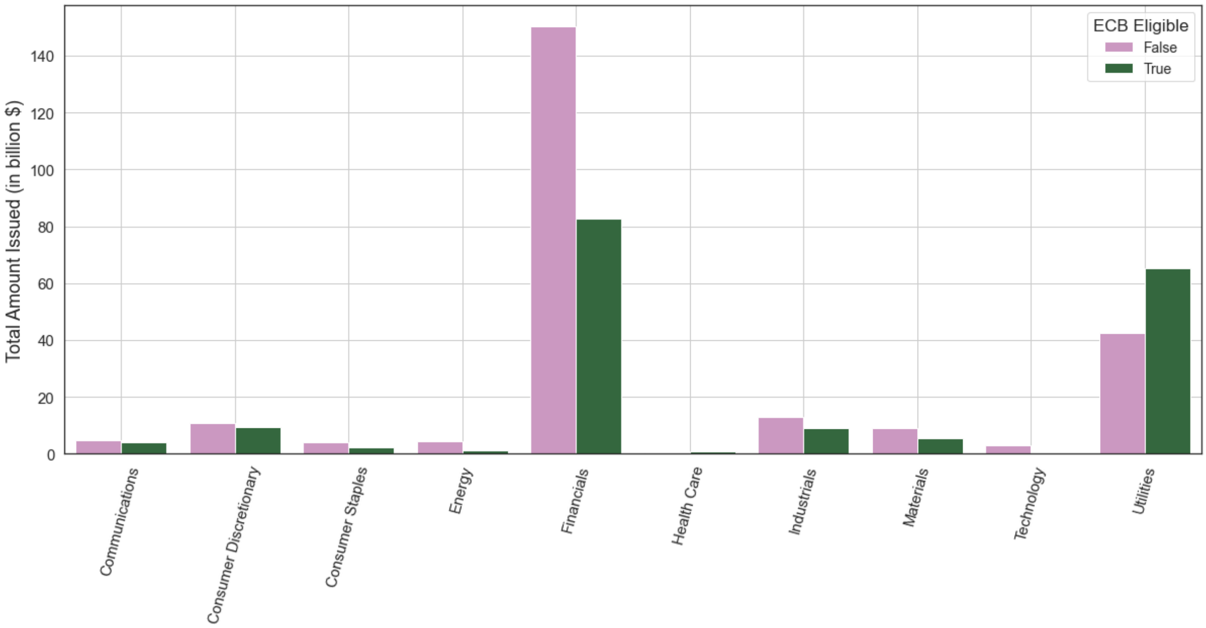


Figure C.5: shows the aggregate amount of EUR-denominated issuance of green bonds (in USD billion) between 2017 and 2021 by BICS Level 1 for ECB-eligible green bonds (green bars) and not-eligible (pink bars) green bonds.

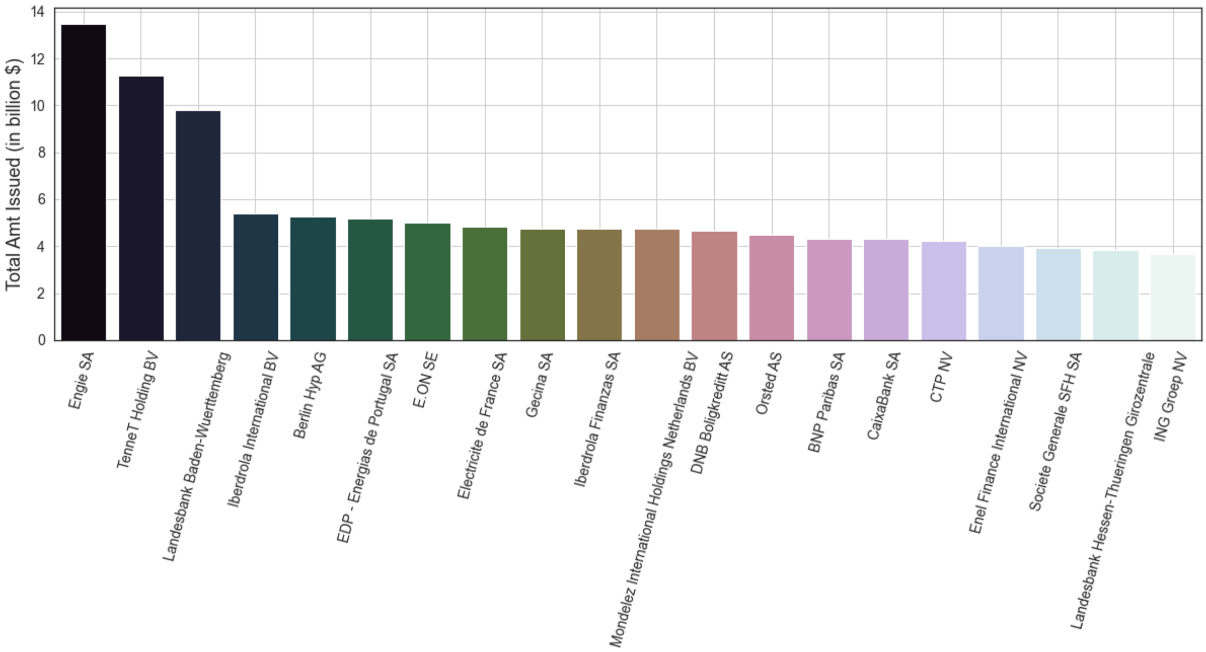


Figure C.6: shows the aggregate amount of EUR-denominated issuance of green bonds (in USD billion) between 2017 and 2021 sorted by top issuing firm.

D Additional Results

D.1 Difference-in-Differences

	All firms				Greenness defined at *sector level*		Greenness defined over *full economy*	
	(1)	(2)	(3)	(4)	(5) Brown Issuers	(6) Non-brown Issuers	(7) Brown Issuers	(8) Non-brown Issuers
Post $\times$*Eligible*	-0.049	-0.061**	-0.048*	-0.057	-0.046***	-0.071	-0.030***	-0.020
	(-0.63)	(-1.99)	(-1.76)	(-1.25)	(-5.80)	(-1.54)	(-8.34)	(-0.35)
Eligible	-0.929***							
	(-3.18)							
Post	-0.056							
	(-0.73)							
Number of distinct bonds	102	102	102	102	31	71	40	62
Adj. R-squared	0.248	0.983	0.986	0.987	0.991	0.988	0.980	0.989
Bond FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Issuer FE								
Country x Month FE	No	No	Yes	No	No	No	No	No
Sector x Month FE	No	No	No	Yes	Yes	Yes	Yes	Yes

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table D.4: This table presents results on the Diff-in-diff on Yields to Maturity after the announcement of the Monetary Policy Strategy Review (on 08/07/2021). Standard errors are clustered at the security (ISIN) level. The treatment group are EUR denominated green bonds issued by non-credit EuroStoxx 600 institutions, while the control group are EUR denominated green bonds issued by non-credit EuroStoxx 600 institutions.

	All firms				Greenness defined over **full economy**		Greenness defined at **sector level**	
	(1)	(2)	(3)	(4)	(5) Brown Issuers	(6) Non-brown Issuers	(7) Brown Issuers	(8) Non-brown Issuers
Post $\times$*Eligible*	0.053***	0.000	0.013	0.015	-0.034***	0.023	-0.058***	0.039***
	(3.93)	(0.06)	(0.71)	(1.19)	(-16.85)	(1.58)	(-4.94)	(2.92)
Eligible	0.081							
	(1.25)							
Post	-0.137***							
	(-19.04)							
Number of distinct bonds	349	349	349	349	37	65	29	73
Adj. R-squared	0.007	0.971	0.976	0.978	0.975	0.978	0.984	0.978
Bond FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country x Month FE	No	No	Yes	No	No	No	No	No
Sector x Month FE	No	No	No	Yes	Yes	Yes	Yes	Yes

t statistics in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table D.5: This table presents results on the Diff-in-diff on Yields to Maturity after the announcement of the Monetary Policy Strategy Review (on 08/07/2021). Standard errors are clustered at the security (ISIN) level. The treatment group are EUR denominated green bonds issued by non-credit EuroStoxx 600 institutions, while the control group are EUR denominated green bonds issued by credit EuroStoxx 600 institutions.